SESSION 5 ASSIGNMENT

From root cause to a testable concept

Complete the plates using evidence from Sessions 1–4 — Assignments 1–3 (NutriScore Checkout Tool, NUT-04).

ROOT CAUSE | VALIDATE | HMW | OPPORTUNITY | DIVERGE | CONVERGE | CONCEPT | VALUE

INDIVIDUAL WORK

1. Complete your own 5 Whys.
2. Complete your own Crazy 8s.
3. Bring all 8 ideas into team convergence.

TEAM WORK

1. Validate / revise the problem frame.
2. Agree on one HMW + opportunity frame.
3. Compare concepts and converge on one direction.
4. Complete the concept card + value proposition.

SUBMIT AS University_TeamName_Session5_Assignment.pdf **Use evidence — not preference.**

5 Whys: from symptom to root cause

Problem symptom

Segment-A shoppers (already managing a diet-sensitive condition) add items to their online cart that turn out not to fit their sugar, salt or fat limits.

Why 1

Because at the moment they choose a product, they cannot tell whether it fits their limits.

Why 2

Because no Kenyan online retailer displays a front-of-pack nutrition signal (Food-EPI Kenya, 2020).

Why 3

Because on-pack labels are too incomplete to work it out themselves — added sugar isn't even a mandatory label field (ATNi / EAMA Kenya Market Assessment, 2025).

Why 4

Because, until NutriScore, nothing cross-referenced Kenyan products against clinical nutrition thresholds and surfaced a grade at the moment of choice.

Why 5 / root cause

Nutrition-quality data and the online purchase decision live in two disconnected systems in Kenya — nothing attaches an evidence-based grade to a specific product at checkout.

Finish with a root cause that can be investigated or designed around.

Problem Validation Decision

Original problem statement

Adults in Kenya who must limit sugar, salt or fat cannot see, when they add an item to an online grocery cart, whether it fits their limits, because retailers show no front-of-pack signal and labels omit fields such as added sugar.

Evidence that supports it

23.8% of Kenyan adults have raised blood pressure and 27.9% are overweight or obese (STEPS 2015); no Kenyan online retailer shows a front-of-pack signal (Food-EPI 2020); no Chrome extension scores a Kenyan grocery site (June 2026 competitive scan); added sugar isn't a mandatory label field (ATNi / EAMA 2025).

Evidence that challenges it

DESK EVIDENCE ONLY

No field interviews conducted yet — everything above is desk evidence. Two challenges from our own analysis: the larger "unaware majority" segment shows low felt urgency (Assignment 2), and only 42–46% of the catalogue can currently show a grade, which could blunt real impact even where the problem is confirmed.

Validation decision: keep / narrow / reframe / park

PARTIAL

KEEP, narrowed to Segment A — partially validated by desk evidence; full validation is pending the field interviews planned in Assignments 1 and 3.

Revised problem statement

Segment-A shoppers cannot tell, at the exact moment they evaluate a specific product online, whether it fits their sugar, salt or fat limits — forcing them to guess, skip categories, or lean on incomplete labels, since no Kenyan retailer shows a front-of-pack signal.

Use this before generating solution concepts.

How Might We Question Plate

Root cause / insight

Nutrition-quality data and the online purchase decision are disconnected systems in Kenya.

User group

Segment A — adults advised to limit sugar, salt or fat, who shop groceries online weekly.

Desired change

They can tell, at a glance, whether a specific product fits their limits — right when they're deciding whether to add it.

HMW option 1 — selected for ideation (slide 6)

SELECTED

How might we help Segment-A shoppers instantly judge whether a product fits their limits, at the exact moment they're deciding to add it to their cart?

HMW option 2

How might we turn Kenya's incomplete on-pack labels into a trustworthy go/no-go signal shoppers can act on without reading every line?

HMW option 3

How might we make "does this fit me" as fast to answer online as it is slow and uncertain in a physical aisle?

A good HMW should not contain your product idea.

Opportunity Frame Plate

For

specific user / segment

Segment-A online grocery shoppers in Kenya — adults advised to limit sugar, salt or fat, who shop weekly.

Who experience

evidence-backed barrier or pain

Not being able to tell, at the moment of choosing a specific product, whether it fits their limits — no Kenyan retailer shows a signal, and labels omit fields like added sugar (Food-EPI 2020; ATNi/EAMA 2025).

Because

root cause / constraint

Nutrition-quality data and the online purchase decision are disconnected systems — nothing currently attaches an evidence-based grade to a specific product at the point of choice.

We have an opportunity to

desired change / outcome

Surface an instant, trustworthy fit signal at the exact moment a shopper evaluates a product — not just a general health score.

Without

ethical, operational or strategic boundary

requiring the shopper to leave their shopping flow or create a new account, or claiming certainty beyond what the catalogue supports — only 42–46% of SKUs currently carry a verified grade.

Use "without" to protect against risky or unrealistic concepts.

Crazy 8s Sketch Plate

HMW question: *How might we help Segment-A shoppers instantly judge whether a product fits their limits, at the exact moment they're deciding to add it to their cart?*

1
Inline fit badge

Colour-coded grade shown directly on the product tile in search and category results.

2
Compare-while-choosing

Show 2–3 same-category alternatives that fit better, right when a low-grade product is opened.

3
Cart-exit check

One pass over the whole cart just before checkout, flagging items that don't fit with a one-tap swap.

4
Photo label check

Shopper photographs a physical pack for items without a product page; gets an instant verdict.

5
Weekly fit digest

A short after-order summary of how many items fit vs didn't — builds awareness over time.

6
SMS / USSD check

A non-digital channel: check one product by barcode or name over SMS, for low-bandwidth shoppers.

7
In-store shelf-talker

A physical tag carrying the same grade in the retailer's physical branches.

8
Clinician safe-list

A dietitian pre-builds a "safe list" that pre-filters a patient's search results.

Rule: no repeating the same idea eight times. Force variation.

Alternative Solution Concepts Plate

Concept 1

Inline Fit Badge

Mechanism
Client-side grade badge on every product tile.

Who
All Segment-A shoppers.

Why it might work
Already shipped and reaches 42–46% of the catalogue — fastest to extend.

Concept 2

CONVERGED

Smart Compare & Swap

Mechanism
At the evaluation moment, surface 2–3 better-fitting alternatives.

Who
Shoppers who see a low grade and want a next step.

Why it might work
Directly targets the journey map's flagged peak-friction stage.

Concept 3

Cart-Exit Guardrail

Mechanism
One consolidated fit-check pass just before checkout.

Who
Time-pressed shoppers who skim product tiles.

Why it might work
Catches what the badge-only flow might get skimmed past.

Concept 4

Clinician Safe-List

Mechanism
A dietitian-built, pre-filtered shopping list shared with a patient.

Who
Newly diagnosed or high-risk patients.

Why it might work
Taps the clinician/KNDI trust relationship already named as a top stakeholder.

Concept 5

SMS / USSD Fit-Check

Mechanism
Barcode or name lookup over SMS, no app or browser needed.

Who
Shoppers with limited data or older phones.

Why it might work
Non-digital option — extends reach beyond desktop-Chrome households.

Concept 6

Photo Label Check

Mechanism
Snap a pack photo for an instant verdict.

Who
Shoppers buying unlisted or new products.

Why it might work
Covers the 54% of the catalogue still unverified.

Try to include at least one non-digital concept.

Idea Convergence Matrix

Concept	Desirable	Feasible	Viable	Responsible	Decision
Inline Fit Badge	4	5	3	4	Proceed — already shipped, extend coverage
Smart Compare & Swap	5	3	3	4	**Priority next build**
Cart-Exit Guardrail	3	4	3	4	Backlog
Clinician Safe-List	4	2	2	5	Park pending KNDI engagement
SMS / USSD Fit-Check	3	2	2	4	Park — needs telco integration
Photo Label Check	3	2	2	3	Park — needs an image/OCR pipeline

Desirable: does Segment A want this? Feasible: can we build it on the shipped architecture? Viable: is there a path to sustain it? Responsible: privacy, trust and safety risk.

Score lightly. Discuss deeply. 1–5 scores only structure the conversation — the evidence behind each score matters more.

Concept Card

Concept name

Smart Compare & Swap

Problem addressed

At the moment a Segment-A shopper evaluates a product, they can see (via the existing badge) that it doesn't fit — but have no fast way to find one that does, so they skip the category or guess.

Why it might work

It completes the pipeline NutriScore already designed but never shipped — the "Suggest" step was spec'd but never coded; the category tagging and scoring it needs already exist in the shipped build.

Smallest test

NEXT VALIDATION STEP

A 1-week moderated pilot with 3–5 Segment-A participants: show a clickable prototype of the compare panel on 10 real product pages (mixing verified and unverified items), and watch whether they use it, ignore it, or hit an empty result.

Priority user

Segment-A shoppers who already see a badge grade but need a next step at the point of choice.

How it works

When a product's grade is C–E for the shopper's flagged concern, the extension surfaces 2–3 same-category products with a better grade, ranked by closeness of fit, pulled from the existing 10,518-SKU dataset — no new data source required.

Key risk

Only 42–46% of the catalogue can currently show a grade, so a fitting alternative may not always exist in the verified portion — the suggestion could come up empty exactly when it's needed most.

Keep the concept testable. Do not overbuild it.

Value Proposition Plate

For
priority user

Segment-A online grocery shoppers in Kenya — adults managing a diet-sensitive condition.

Who struggle with
validated pain / job / barrier

Knowing a product doesn't fit their limits but having no fast way to find one that does, at the exact moment they're choosing.

Our concept helps them
desired outcome

See a better-fitting alternative in the same category, right where they're already deciding — without leaving the page or starting a new search.

Unlike current alternatives
current workaround or competitor

Today they guess, memorise avoid-lists, or abandon the category — no Kenyan retailer or app surfaces an alternative automatically (Food-EPI 2020; June 2026 competitive scan).

Because
evidence-backed advantage

It reuses the grading and category data NutriScore already computes for 10,518 products — no new data source or manual curation is required to launch.

We will know it works if
test signal or behaviour change

In the pilot, at least half of shoppers shown a low-grade product tap through to a suggested alternative, and say it saved them a step they'd otherwise do themselves.

The proposition should be clear before the team designs screens or features.